LLMs Meet Time Series Models: Harnessing Internal Knowledge for Few-Shot Forecast of Migration Flows

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Background

- **Motivation:** **Accurate short-term forecasts are essential** for responding to urgent situations like migration surges where timely preparation is critical
- **Problem:** **Sparse and short time series** data make forecasting difficult
- **Limitation:** Traditional models **struggle to identify** which exogeneous variables meaningfully affect the target
- **Idea:** We propose using LLM to select the most impactful exogenous variables using **external knowledge**, improving accuracy and reducing noise when data is limited

Exogenous Variables
- ➢ US CPI
- ➢ India GDP
- ❑ Pressure
- ❑ Rainfall

→

Target Variable
- ➢ Migration to Western Asia
- ❑ Outside Temperature

- **Question:** How do we know which exogenous variables are the most important to predict the target variable?
- **Solution:** Large language models (LLMs) have contextual abilities to determine relationships without prior knowledge

Methodology

Datasets

5 datasets contain **monthly migration counts and economic factors (GDP, CPI, etc.)** from countries in Europe, Asia, and Africa

We used **11 datasets** across a variety of domains

6 datasets are **common multivariate time series datasets from environmental and electricity domains** with varying frequencies

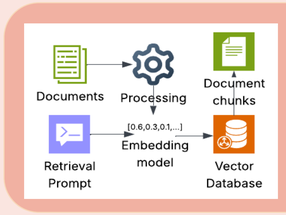
Retreival Augmented Generation

Feature Selection Methods

We query the LLM to select n exogenous variables 10 times and take the n most selected variables from the 10 queries

We compare our LLM selection with:
- Random feature selection
- Recursive feature elimination
- Importance scores from tree ensembles

Data → **Target Variable** / **Exogenous Variables** → **Prompt Generation** → **LLM** → **Top k Selected Exogenous Variables** → **Time Series Model** → **Forecast**



Prompt Design & RAG

1. RAG context from document chunks
2. System prompt for LLM role as forecaster
3. User Prompt: Select and Rank n exogenous variables:
   - Consider causal relevance, predictive power, and lead/lag relationships

We used **2 LLMs** to select exogenous variables
- GPT-4o-mini
- Gemini's Flash 2.0

Time Series Models

Traditional Time Series Models
- SARIMAX
- Prophet

Fine-tuned both models for each dataset

We used **5 time series models for forecasting** including both traditional and foundational models.

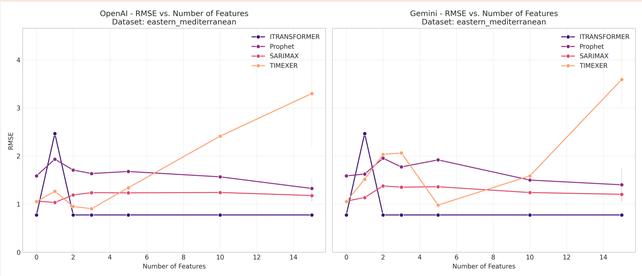
Time Series Foundational Models
- TimeXer
- iTransformer

Pre-trained on large amounts of time series data

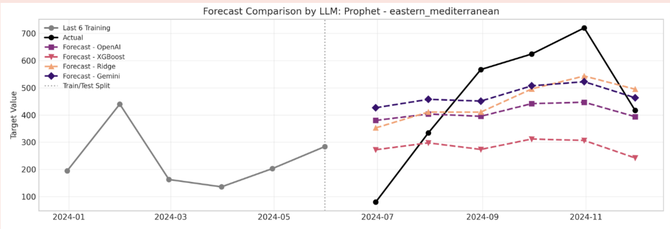
Research Question

Can we use large language models to extract information about relationships between exogenous variables and target variables for time series forecasting of migration flows?

Results



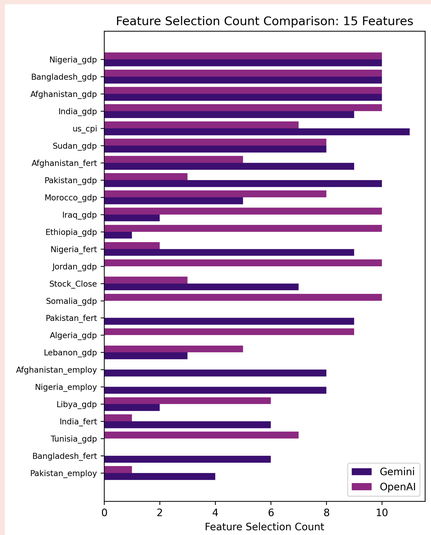
Top: RSME for Eastern Mediterranean dataset over different numbers of features for each LLM and model
Bottom: Forecast comparison of different feature selection methods for Eastern Mediterranean dataset using Prophet model



Normalized MAE for all migration datasets

Normalized MAE

Dataset	Method	Model	0	1	2	3	5	10	15	157
Central Mediterranean	SARIMAX	Gemini	0.70	**0.47**	1.07	1.25	1.49	1.35	1.68	7.74
		OpenAI	0.70	0.47	0.66	0.79	0.90	0.68	**0.29**	**7.74**
	Prophet	Gemini	1.39	**0.93**	2.63	2.72	2.31	1.19	1.78	0.95
		OpenAI	1.39	1.29	1.13	1.04	1.01	1.90	1.47	**0.95**
	TimerXer	Gemini	0.62	0.68	0.62	0.39	0.36	0.48	**0.22**	0.53
		OpenAI	0.62	0.47	0.67	0.54	0.36	0.40	**0.29**	**0.53**
	iTransformer	Gemini	0.32	**0.26**	0.32	0.32	0.32	0.32	0.32	0.32
		OpenAI	0.32	**0.26**	0.32	0.32	0.32	0.32	0.32	0.32
Eastern Borders	SARIMAX	Gemini	**1.95**	2.08	2.02	2.05	2.12	2.13	2.18	14.13
		OpenAI	1.95	2.16	2.10	2.08	2.14	2.08	**0.35**	14.13
	Prophet	Gemini	0.89	0.85	0.71	0.71	0.59	0.49	0.43	**0.35**
		OpenAI	0.89	0.82	0.80	0.70	0.67	0.54	0.42	**0.35**
	TimerXer	Gemini	**0.42**	0.45	0.99	0.77	0.79	0.73	0.86	0.82
		OpenAI	**0.42**	0.66	0.56	0.67	0.75	0.79	0.81	0.82
	iTransformer	Gemini	1.06	**0.70**	1.06	1.06	1.06	1.06	1.06	1.06
		OpenAI	1.06	**0.70**	1.06	1.06	1.06	1.06	1.06	1.06
Eastern Mediterranean	SARIMAX	Gemini	0.86	**0.75**	0.76	0.75	0.75	0.78	0.80	3.53
		OpenAI	0.86	0.89	**0.69**	0.69	0.73	0.77	0.77	3.53
	Prophet	Gemini	0.83	0.85	0.99	0.91	0.97	0.83	0.90	**0.65**
		OpenAI	0.83	0.98	0.89	0.85	0.87	0.84	**0.82**	**0.65**
	TimerXer	Gemini	0.71	0.87	1.10	1.05	**0.66**	1.00	1.62	1.93
		OpenAI	**0.71**	0.93	0.84	0.80	0.96	1.39	1.28	1.93
	iTransformer	Gemini	**0.77**	1.77	0.77	0.77	0.77	0.77	0.77	0.77
		OpenAI	**0.77**	1.77	0.77	0.77	0.77	0.77	0.77	0.77
Western Balkan	SARIMAX	Gemini	**0.62**	3.47	4.11	5.34	0.79	2.51	11.56	23.85
		OpenAI	**0.62**	0.73	2.58	0.71	1.11	3.18	4.09	23.85
	Prophet	Gemini	6.69	9.66	8.15	8.32	4.12	4.55	8.47	**1.07**
		OpenAI	6.69	5.27	1.29	3.47	4.65	5.99	1.13	**1.07**
	TimerXer	Gemini	**2.37**	5.80	29.84	18.35	17.68	6.02	15.32	12.08
		OpenAI	**2.37**	6.98	11.46	16.51	9.28	3.88	4.84	12.08
	iTransformer	Gemini	5.16	**2.16**	5.16	5.16	5.16	5.16	5.16	5.16
		OpenAI	5.16	**2.16**	5.16	5.16	5.16	5.16	5.16	5.16
Western Mediterranean	SARIMAX	Gemini	**0.48**	3.33	1.64	2.13	1.34	2.65	1.44	38.35
		OpenAI	**0.48**	1.15	2.66	0.89	0.89	3.81	0.66	38.35
	Prophet	Gemini	5.60	4.02	3.32	3.47	**2.25**	2.54	2.45	8.54
		OpenAI	5.60	4.12	4.09	3.49	3.38	**2.47**	2.77	8.54
	TimerXer	Gemini	0.93	1.54	**0.67**	0.70	0.71	0.75	0.80	1.71
		OpenAI	0.93	0.85	0.71	0.79	0.67	0.67	**0.60**	1.71
	iTransformer	Gemini	1.18	**0.66**	1.18	1.18	1.18	1.18	1.18	1.18
		OpenAI	1.18	**0.66**	1.18	1.18	1.18	1.18	1.18	1.18



Features selected by each LLM for the Eastern Mediterranean dataset when prompted for 15 features

Discussion

- **Takeaway:** Selecting a subset of exogenous features is mostly better than using all exogenous variables
- **Takeaway:** In some cases, LLMs do a good or better job than other feature selection methods
- **Takeaway:** It is difficult to determine how many exogenous variables to select – very dataset and model specific
- **Future Work:** Could have the LLM encode information about how important each variable is to the target variable, rather than select features

References and Acknowledgements

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For a complete list of references and acknowledgments, please scan here!